=== UNIT 6: SOLID STATE PHYSICS, CRYSTALLOGRAPHY & BAND THEORY ===

>> 6.1 Crystal Geometry, Lattices, and Unit Cells

Solid state physics studies the microscopic atomic architecture and macroscopic physical properties of condensed matter. In civil materials science, crystal structure governs the strength of steel rebar, hydration of Portland cement, and stability of soil clay minerals.
* Crystal Lattice: An infinite periodic mathematical array of geometric points in 3D space, where every lattice point has surroundings identical to every other point.
* Basis (Motif): An atom, ion, or group of atoms attached identically to every lattice point: Crystal Structure = Lattice + Basis.
* Unit Cell: The fundamental repeating volumetric building block of a crystal that reproduces the entire crystal lattice upon repeated spatial translations by primitive vectors a, b, c.
* Primitive Cell: A minimum-volume unit cell containing strictly ONE lattice point.
* Bravais Lattices: In 3D space, exactly 14 unique Bravais lattices exist, categorized into Seven Crystal Systems: (1) Cubic, (2) Tetragonal, (3) Orthorhombic, (4) Rhombohedral (Trigonal), (5) Hexagonal, (6) Monoclinic, (7) Triclinic.

>> 6.2 Cubic Crystal Structures: SC, BCC, and FCC

* 1. Simple Cubic (SC): Lattice points at 8 corners. Total atoms per cell: N = 8 * (1/8) = 1. Coordination Number = 6. Nearest neighbor distance: 2*r = a. Atomic Packing Factor: APF = pi / 6 ~= 0.52 (52%).
* 2. Body-Centered Cubic (BCC): Points at 8 corners + 1 at body center (e.g., alpha-Ferrite steel at room temp). Atoms per cell: N = 8*(1/8) + 1 = 2. Coordination Number = 8. Body diagonal: 4*r = a * sqrt(3) => r = a * sqrt(3) / 4. APF = pi * sqrt(3) / 8 ~= 0.68 (68%).
* 3. Face-Centered Cubic (FCC): Points at 8 corners + 6 face centers (e.g., Austenite steel, Copper, Aluminum). Atoms per cell: N = 8*(1/8) + 6*(1/2) = 4. Coordination Number = 12. Face diagonal: 4*r = a * sqrt(2) => r = a * sqrt(2) / 4. APF = pi * sqrt(2) / 6 ~= 0.74 (74%, close-packed).

>> 6.3 Miller Indices of Crystallographic Planes

Miller indices (hkl) uniquely define orientation of crystal lattice planes:
* Determination Algorithm:
- Step 1: Determine intercepts of the plane along crystallographic axes in terms of lattice constants: (p*a, q*b, r*c).
- Step 2: Take reciprocals of intercept multipliers: (1/p, 1/q, 1/r). (If plane is parallel to an axis, intercept is infinity => reciprocal is 0).

- Step 3: Clear fractions by multiplying by least common denominator (LCD) to obtain smallest integers: (h, k, l).
- Step 4: Enclose in parentheses without commas: (h k l). Negative intercepts indicated by a bar over index: (h bar-k l).
* Interplanar Spacing (d_hkl): Perpendicular distance between adjacent parallel planes in a cubic crystal of lattice constant a:
Formula: d_hkl = a / sqrt(h^2 + k^2 + l^2).

>> 6.4 X-Ray Diffraction and Bragg's Law

X-rays have wavelengths (lambda ~ 0.1 nm) comparable to interatomic lattice spacings, producing coherent constructive diffraction interference:
* Bragg's Law of Diffraction: Consider two parallel lattice planes separated by distance d. Path difference between rays reflected from adjacent planes is 2 * d * sin(theta):
Constructive Interference Condition: 2 * d * sin(theta) = n * lambda, where theta is Bragg glancing angle and n is diffraction order (n = 1, 2, ...).
* Powder Method: Monochromatic X-rays incident on powdered polycrystalline sample produce concentric diffraction cones recorded on Debye-Scherrer photographic film or diffractometer detector.

>> 6.5 Free Electron Theory of Metals

* 1. Classical Drude-Lorentz Model: Assumed valence electrons form an ideal classical gas obeying Maxwell-Boltzmann statistics. Successfully explained Ohm's law, but failed catastrophically to predict electronic specific heat and Wiedemann-Franz ratio at low temperatures.
* 2. Quantum Sommerfeld Free Electron Theory: Incorporates Pauli Exclusion Principle and Fermi-Dirac Statistics:
- Fermi-Dirac Distribution Function: Probability f(E) that a quantum state of energy E is occupied by an electron at temperature T:
Formula: f(E) = 1 / [e^((E - E_F) / (k_B * T)) + 1], where E_F is Fermi Energy.
- At absolute zero (T = 0 K): For E < E_F: f(E) = 1.0 (all states filled). For E > E_F: f(E) = 0.0 (all states empty).
- Fermi Energy at 0 K in 3D: E_F0 = (h_bar^2 / 2m) * (3 * pi^2 * n)^(2/3), where n = N / V is electron density (m^-3).
- Fermi Velocity: v_F = sqrt(2 * E_F / m). Fermi Temperature: T_F = E_F / k_B.

>> 6.6 Band Theory of Solids and Classification of Materials

In isolated atoms, electrons occupy sharp discrete energy levels. When atoms condense into an ordered crystal lattice, overlapping atomic wave functions and periodic electrostatic potentials split discrete levels into continuous Energy Bands:

* Kronig-Penney Model: Solves Schrodinger equation for 1D periodic square-well potential representing crystalline lattice ion cores. Demonstrates mathematically that electron energy spectrum divides into Allowed Energy Bands and Forbidden Energy Gaps (Bandgap E_g).
* Material Classification based on Band Structure:
- 1. Conductors (Metals): Valence band and conduction band overlap, or conduction band is partially filled. Zero bandgap (E_g = 0). Electrons easily accelerate into empty states under tiny applied electric fields.
- 2. Semiconductors: Completely filled valence band separated from empty conduction band by a small forbidden energy bandgap (E_g ~ 0.5 to 2.5 eV, e.g. Silicon E_g = 1.1 eV). Thermal excitation at room temperature kicks electrons across bandgap, creating conduction electrons and mobile holes.
- 3. Insulators: Large forbidden bandgap (E_g > 5.0 eV, e.g. Diamond E_g = 5.4 eV). Thermal energy is insufficient to bridge gap; electrical conductivity is negligible.
* Concept of Effective Mass (m*): Incorporates the dynamic influence of internal periodic crystalline lattice forces into electron acceleration under external field: m* = h_bar^2 / (d^2E / dk^2).

[EXAMPLE 1: Bragg's Law X-Ray Diffraction on Copper FCC Crystal]

Problem Context: Copper crystallizes in an FCC lattice. An X-ray diffractometer using Cu-K_alpha radiation of wavelength lambda = 0.1542 nm (1.542 Angstroms) records the first-order (n = 1) reflection peak from the (1 1 1) planes at glancing Bragg angle theta = 21.65 deg. Calculate: (1) Interplanar spacing d_111, (2) Lattice parameter a of copper, (3) Atomic radius r of copper, (4) Density of copper (Atomic weight = 63.55 g/mol, N_A = 6.022 * 10^23 mol^-1).
Solution Step 1: Compute Interplanar Spacing d_111 using Bragg's Law:
2 * d * sin(theta) = n * lambda => d_111 = (1 * 0.1542 nm) / (2 * sin(21.65)).
sin(21.65 deg) = 0.3689.
d_111 = 0.1542 / (2 * 0.3689) = 0.1542 / 0.7378 = 0.2090 nm (2.090 Angstroms).
Solution Step 2: Compute Lattice Parameter a:
d_hkl = a / sqrt(h^2 + k^2 + l^2) => a = d_111 * sqrt(1^2 + 1^2 + 1^2) = d_111 * sqrt(3).
a = 0.2090 nm * 1.73205 = 0.3620 nm = 3.620 * 10^-10 m (3.620 Angstroms).
Solution Step 3: Compute Atomic Radius r for FCC (4*r = a*sqrt(2)):
r = a * sqrt(2) / 4 = 0.3620 nm * 1.4142 / 4 = 0.1280 nm = 1.280 Angstroms.
Solution Step 4: Compute Theoretical Mass Density rho:
FCC unit cell contains n_atoms = 4 atoms.
Unit cell volume V_cell = a^3 = (3.620 * 10^-8 cm)^3 = 4.7438 * 10^-23 cm^3.
Mass of unit cell = (n_atoms * Atomic Weight) / N_A = (4 * 63.55 g) / (6.022 * 10^23) = 4.221 * 10^-22 g.
rho = Mass / Volume = 4.221 * 10^-22 g / (4.7438 * 10^-23 cm^3) = 8.898 g/cm^3 = 8898 kg/m^3 (matches empirical value of 8900 kg/m^3).
Conclusion: d_111 = 0.209 nm; lattice parameter a = 0.362 nm; atomic radius r = 0.128

nm; density = 8898 kg/m^3.

[EXAMPLE 2: Determination of Miller Indices of Crystallographic Planes]

Problem Context: A crystallographic plane in a simple orthorhombic lattice has intercepts at 2a along x-axis, 3b along y-axis, and is parallel to the z-axis (intercept = infinity). A second plane intersects axes at 3a, -2b, and 1c. Determine the Miller indices for both planes.
Solution Step 1: Plane 1 Analysis:
Intercepts: (2a, 3b, infinity*c).
Divide by unit cell dimensions: (2, 3, infinity).
Take reciprocals: (1/2, 1/3, 1/infinity) = (1/2, 1/3, 0).
Multiply by LCM (6): (3, 2, 0).
Miller Indices for Plane 1: (3 2 0).
Solution Step 2: Plane 2 Analysis:
Intercepts: (3a, -2b, 1c).
Divide by unit cell dimensions: (3, -2, 1).
Take reciprocals: (1/3, -1/2, 1/1).
Multiply by LCM (6): (2, -3, 6).
Represent negative index with bar: (2 bar-3 6).
Miller Indices for Plane 2: (2 3_bar 6).
Conclusion: Plane 1 = (3 2 0); Plane 2 = (2 -3 6) or (2 3_bar 6).

[EXAMPLE 3: Fermi Energy, Velocity and Temperature in Copper]

Problem Context: Copper is monovalent with one free conduction electron per atom. Free electron density is n = 8.49 * 10^28 m^-3. Calculate: (1) Fermi Energy E_F at 0 K in Joules and electron-volts, (2) Fermi Velocity v_F of electrons at Fermi surface, (3) Fermi Temperature T_F.
Solution Step 1: Compute Fermi Energy E_F:
E_F = (h_bar^2 / 2*m_e) * (3 * pi^2 * n)^(2/3).
h_bar = 1.05457 * 10^-34 J-s; m_e = 9.109 * 10^-31 kg.
Constant term: h_bar^2 / (2 * m_e) = (1.05457 * 10^-34)^2 / (2 * 9.109 * 10^-31) = 6.104 * 10^-39 J-m^2.
3 * pi^2 * n = 3 * (3.14159)^2 * 8.49 * 10^28 = 29.6088 * 8.49 * 10^28 = 2.5138 * 10^30 m^-3.
(2.5138 * 10^30)^(2/3) = [(2.5138)^(2/3)] * 10^20 = 1.8488 * 10^20 m^-2.
E_F = (6.104 * 10^-39) * (1.8488 * 10^20) = 1.1285 * 10^-18 Joules.
In eV: E_F = 1.1285 * 10^-18 J / (1.6022 * 10^-19 J/eV) = 7.043 eV.
Solution Step 2: Compute Fermi Velocity v_F:
0.5 * m_e * v_F^2 = E_F => v_F = sqrt(2 * E_F / m_e).
v_F = sqrt(2 * 1.1285 * 10^-18 / 9.109 * 10^-31) = sqrt(2.4777 * 10^12) = 1.574 *

10^6 m/s (~ 1574 km/s).
Solution Step 3: Compute Fermi Temperature T_F:
T_F = E_F / k_B = 1.1285 * 10^-18 J / (1.3806 * 10^-23 J/K) = 81,740 Kelvin.
Conclusion: Fermi Energy E_F = 7.04 eV; Fermi Velocity v_F = 1.57 * 10^6 m/s; Fermi Temperature T_F = 81,740 K.

[EXAMPLE 4: Thermal Electron Occupancy Probability via Fermi-Dirac]

Problem Context: For a metallic conductor at temperature T = 300 K (room temp, k_B*T = 0.02586 eV), calculate: (1) Probability of an electron occupying an energy state at E = E_F + 0.05 eV, (2) Probability of an empty state (hole) at E = E_F - 0.05 eV, (3) Occupancy probability at E = E_F + 0.05 eV if temperature rises to T = 1200 K (k_B*T = 0.10344 eV).
Solution Step 1: Occupancy at E = E_F + 0.05 eV at T = 300 K:
Energy difference Delta E = E - E_F = + 0.05 eV.
Exponent: (E - E_F) / (k_B * T) = 0.05 eV / 0.02586 eV = 1.9335.
f(E) = 1 / [e^(1.9335) + 1] = 1 / [6.9137 + 1] = 1 / 7.9137 = 0.1264 (12.64%).
Solution Step 2: Probability of Hole (Unoccupied state) at E = E_F - 0.05 eV at T = 300 K:
Delta E = E - E_F = - 0.05 eV => Exponent = - 1.9335.
f(E) = 1 / [e^(-1.9335) + 1] = 1 / [0.1446 + 1] = 1 / 1.1446 = 0.8736.
Probability of hole = 1 - f(E) = 1 - 0.8736 = 0.1264 (12.64%, exactly symmetrical about E_F!).
Solution Step 3: Occupancy at E = E_F + 0.05 eV at T = 1200 K:
Exponent = 0.05 eV / 0.10344 eV = 0.4834.
f(E) = 1 / [e^(0.4834) + 1] = 1 / [1.6216 + 1] = 1 / 2.6216 = 0.3814 (38.14%).
Conclusion: At 300 K, occupancy probability at E_F + 0.05 eV is 12.64%; at 1200 K, thermal agitation elevates occupancy probability to 38.14%.

>> 6.7 University Review & Examination Questions

* Q1: Derive Bragg’s Law of X-ray diffraction and explain how it is utilized to deduce crystal structures experimentally.
Answer: Consider parallel crystallographic planes separated by spacing d. Parallel monochromatic X-ray beams 1 and 2 strike adjacent planes at glancing angle theta. Ray 2 penetrates to the second plane and is specularly reflected. Dropping perpendiculars from incidence point A onto ray 2 at B and C, the extra path traversed by ray 2 is Delta = AB + AC = d*sin(theta) + d*sin(theta) = 2*d*sin(theta). For constructive interference, this path difference must equal an integral multiple of wavelength: 2*d*sin(theta) = n*lambda (Bragg’s Law). In experimental crystallography (X-ray diffractometer), a sample is illuminated with known X-ray wavelength lambda while detector scans angles 2*theta. Measuring diffraction peak positions theta directly reveals interplanar spacings d_hkl = n*lambda / (2*sin(theta)). Matching observed d_hkl values against quadratic forms of

crystal systems reveals lattice type (SC, BCC, FCC) and lattice parameter a.
* Q2: Explain the Kronig-Penney model and illustrate how it explains the formation of allowed and forbidden energy bands in solids.
Answer: The Kronig-Penney model idealizes the 1D periodic potential of a crystalline lattice as a periodic array of rectangular potential wells and barriers of period (a + b), where well width is a (ion core) and barrier width is b with height V0. Solving the 1D Schrodinger equation in both regions and applying Bloch theorem (psi(x) = u_k(x)*e^(ikx)) with boundary continuity leads to the dispersion relation: P * [sin(alpha*a) / (alpha*a)] + cos(alpha*a) = cos(k*a), where P = (m*V0*b*a) / h_bar^2 is barrier strength and alpha = sqrt(2mE)/h_bar. Since cos(ka) can only assume values between -1 and +1, only those electron energies E for which the left-hand function lies within [-1, +1] are permitted. Values where the function exceeds [-1, +1] are forbidden, forming Forbidden Energy Bandgaps. This explains fundamentally why solids split into conductors, semiconductors, and insulators.

>> 6.8 Key Takeaways and Revision Points

* Bragg’s law 2*d*sin(theta) = n*lambda governs X-ray crystal diffraction.
* Interplanar spacing for cubic lattice: d_hkl = a / sqrt(h^2 + k^2 + l^2).
* Fermi energy E_F represents highest filled electron energy level at absolute zero (T = 0 K).
* Band theory explains conductivity: Metals have overlapping bands; semiconductors have small gap (E_g ~ 1 eV); insulators have wide gap (E_g > 5 eV).